

Bachelor Applied Data Science & Artificial Intelligence

Course Name	Calculus
Course Code	PR12
Performance Indicator	1
Academic Year	2025–2026
Block period	4
Test / Resit	Mock Test
Test Date	N/A
Start Time – End Time	N/A
Separate answer sheet	No
Examiners	Baljan, J. & Alberts, F.
Maximum Number of Points (T)	40
Randomly Expected Points (E)	0
Passing Threshold Points	22

Number of Points (P)	
Student Name	
Student Identification Number	

Calculation PI-grade: $(P-E)/(T-E)*10$.

The examination board can make a generic correction to this after test analysis.

Instructions

- Allowed aids: writing utensils, pen with black or blue ink
- Check the entire set for completeness and call the exam supervisor if parts are missing
- Complete the entries in the spaces indicated on this form and/or on the answer sheet
- Hand everything in to the exam supervisor afterwards.

Question 1 (2 points)

Consider the function

$$f(x) = \begin{cases} \frac{x^2 - 4x}{x - 4}, & x \neq 4, \\ a, & x = 4. \end{cases}$$

Determine the value of a for which f is continuous at $x = 4$.

For $x \neq 4$,

$$\frac{x^2 - 4x}{x - 4} = \frac{x(x - 4)}{x - 4} = x.$$

Hence

$$\lim_{x \rightarrow 4} f(x) = \lim_{x \rightarrow 4} (x) = 4.$$

(1 point)

For continuity at $x = 4$ we need $f(4) = \lim_{x \rightarrow 4} f(x)$, so $a = 4$.

(1 point)

Question 2 (3 points)

Consider the function

$$g(x) = \begin{cases} 3x^2 + b, & x < 2, \\ 6, & x = 2, \\ bx + 18, & x > 2. \end{cases}$$

Determine all values of b for which g is continuous at $x = 2$.

For continuity at $x = 2$, we need

$$\lim_{x \rightarrow 2^-} g(x) = g(2) = \lim_{x \rightarrow 2^+} g(x).$$

The left-hand limit is

$$\lim_{x \rightarrow 2^-} (3x^2 + b) = 12 + b.$$

(1 point)

The right-hand limit is

$$\lim_{x \rightarrow 2^+} (bx + 18) = 2b + 18$$

(1 point)

Since $g(2) = 6$, we need

$$12 + b = 6 \quad \text{and} \quad 2b + 18 = 6.$$

Solving for b gives $b = -6$ in both cases. Therefore,

$$b = -6.$$

(1 point)

Question 3 (4 points)

Consider the function

$$h(x) = \sin(2x)e^{\sin(2x)}$$

Find $h'(x)$

Use the product rule

$$h(x) = f(x)g(x)$$

with

$$f(x) = \sin(2x), \quad g(x) = e^{\sin(2x)}$$

Using the chain rule,

$$f'(x) = 2 \cos(2x)$$

(1 point)

For $g(x)$ we apply the chain rule twice, resulting in

$$g'(x) = 2e^{\sin(2x)} \cos(2x)$$

(2 points)

Applying the product rule, we get:

$$h'(x) = f'(x)g(x) + f(x)g'(x)$$

Thus,

$$\begin{aligned} h'(x) &= 2 \cos(2x)e^{\sin(2x)} + 2 \sin(2x)e^{\sin(2x)} \cos(2x) \\ &= 2 \cos(2x)e^{\sin(2x)}(1 + \sin(2x)) \end{aligned}$$

(1 point)

Question 4 (4 points)

Given the function

$$f(x) = \frac{1}{3}x^3 + x^2 - 8x - 5$$

Find the critical points of f , classify them using the first or second derivative test, and determine the interval where f is concave up and concave down.

First compute

$$f'(x) = x^2 + 2x - 8 = (x + 4)(x - 2).$$

So the critical points are

$$x = -4, \quad x = 2.$$

(1 point)

The second derivative is

$$f''(x) = 2x + 2.$$

At $x = -4$,

$$f''(-4) = -6 < 0,$$

so f has a local maximum at $x = -4$.

(1 point)

At $x = 2$,

$$f''(2) = 6 > 0,$$

so f has a local minimum at $x = 2$.

(1 point)

For convexity, solve

$$f''(x) = 2x + 2 = 0 \implies x = -1.$$

Thus $f''(x) < 0$ when $x < -1$, so f is concave down on $(-\infty, -1)$. Also $f''(x) > 0$ when $x > -1$, so f is concave up on $(-1, \infty)$.

(1 point)

Question 5 (6 points)

Compute

$$\int_0^2 x^2 e^x dx$$

We use integration by parts with

$$u = x^2, \quad dv = e^x dx$$

$$du = 2x dx, \quad v = e^x$$

(1 point)

Therefore,

$$\int x^2 e^x dx = x^2 e^x - 2 \int e^x x dx$$

(1 point)

Using integration by parts again on

$$\int e^x x dx$$

We get

$$u = x, \quad dv = e^x dx$$

$$du = 1 dx, \quad v = e^x$$

(1 point)

Thus,

$$\int e^x x dx = x e^x - \int e^x dx = x e^x - e^x$$

(1 point)

Substituting back

$$\int x^2 e^x dx = x^2 e^x - 2 \int e^x x dx = x^2 e^x - 2(x e^x - e^x) = e^x (x^2 - 2x + 2) + C$$

(1 point)

Now to find the definite integral

$$\begin{aligned} \int_0^2 x^2 e^x dx &= [e^x (x^2 - 2x + 2) + C]_0^2 = \\ &= e^2 (2^2 - 2 \cdot 2 + 2) + C - (e^0 (0 - 0 + 2) + C) = \\ &= 2e^2 - 2 \end{aligned}$$

(1 point)

Question 6 (4 points)

Evaluate

$$\int \frac{3x^2}{x^3 + 1} dx$$

We use substitution:

$$u = x^3 + 1$$

Thus

$$du = 3x^2 dx$$

(1 point)

Therefore

$$\int \frac{3x^2}{x^3 + 1} dx = \int \frac{1}{u} du$$

(1 point)

Hence

$$\int \frac{3x^2}{x^3 + 1} dx = \ln |u| + C$$

(1 point)

Substituting back,

$$\int \frac{3x^2}{x^3 + 1} dx = \ln |x^3 + 1| + C$$

(1 point)

Question 7 (5 points)

Construct the degree 3 Maclaurin polynomial of

$$f(x) = \sin 2x$$

We first find the derivatives

$$f'(x) = 2 \cos 2x$$

$$f''(x) = -4 \sin 2x$$

$$f'''(x) = -8 \cos 2x$$

(3 points)

Evaluate these at $x = 0$:

$$f(0) = 0, \quad f'(0) = 2, \quad f''(0) = 0, \quad f'''(0) = -8$$

(1 point)

The degree 3 Maclaurin polynomial takes the form

$$P_3(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3.$$

Therefore,

$$P_3(x) = 0 + 2x + 0 + \frac{-8}{6}x^3 = 2x + \frac{-4}{3}x^3.$$

(1 point)

Question 8 (6 points)

Solve the initial value problem

$$y' + 4y = e^{-3x}, \quad y(0) = 2$$

This is a first-order linear differential equation:

$$y' + 4y = e^{-3x}.$$

The integrating factor is

$$\mu(x) = e^{\int 4 dx} = e^{4x}.$$

(1 point)

Multiply both sides by e^{4x} :

$$e^{4x}y' + 4e^{4x}y = e^x.$$

The left-hand side is

$$\frac{d}{dx} (e^{4x}y).$$

(2 points)

Thus

$$\frac{d}{dx} (e^{4x}y) = e^x.$$

Integrating gives

$$e^{4x}y = e^x + C.$$

(1 point)

Therefore,

$$y = e^{-3x} + Ce^{-4x}.$$

(1 point)

Using $y(0) = 2$,

$$2 = 1 + C,$$

so

$$C = 1.$$

Hence

$$y = e^{-3x} + e^{-4x}.$$

(1 point)

Question 9 (6 points)

Find the maximum and minimum values of

$$f(x, y) = -2x + 2y$$

subject to the constraint

$$x^2 + y^2 = 3.$$

Let

$$g(x, y) = x^2 + y^2 - 3.$$

Then

$$\nabla f = (-2, 2), \quad \nabla g = (2x, 2y).$$

(1 point)

Using Lagrange multipliers,

$$\nabla f = \lambda \nabla g.$$

So

$$-2 = 2\lambda x, \quad 2 = 2\lambda y, \quad x^2 + y^2 = 3.$$

(1 point)

From the first two equations,

$$x = -\frac{1}{\lambda}, \quad y = \frac{1}{\lambda}.$$

Therefore,

$$y = -x.$$

Substitute this into the constraint:

$$x^2 + (-x)^2 = 3.$$

So

$$2x^2 = 3 \implies x = \pm \sqrt{\frac{3}{2}}.$$

Since $y = -x$ (y depends on x), we get the candidate points

$$\left(\sqrt{\frac{3}{2}}, -\sqrt{\frac{3}{2}} \right), \quad \left(-\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}} \right).$$

(2 point)

Now evaluate f on both candidate points and determine the maximum and minimum:

$$f\left(\sqrt{\frac{3}{2}}, -\sqrt{\frac{3}{2}}\right) = -2\sqrt{\frac{3}{2}} - 2\sqrt{\frac{3}{2}} = -4\sqrt{\frac{3}{2}}$$

Also,

$$f\left(-\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}}\right) = 2\sqrt{\frac{3}{2}} + 2\sqrt{\frac{3}{2}} = 4\sqrt{\frac{3}{2}}$$

Therefore, the maximum value is $\max f = 4\sqrt{\frac{3}{2}}$, and the minimum value is $\min f = -4\sqrt{\frac{3}{2}}$.

(2 point)